MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPES 6H6 AND 12H6
a b

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin diode, metal

Outline --- 5-1

Base --- Phenolic B7-22

Envelope --- MT-8

Cathode --- Coated unipotential

Base connections:

Pin no.	1	2	3	4	5	7	8
Element ---	sh	h	2a	2k	1a	h	1k

ABSOLUTE RATINGS:

	TYPE							
Parameter:	a, b	Ef	epx	Epp/a	Ehk	Esh	ib	Io
Unit:	a, b	V	v	Vac	v	V	ma	mAdc
Maximum:	a	6.9	465	---	360	---	50	8.8
	b	13.9	---	---	---	---	--	---
Minimum:	a	5.7	---	---	---	---	--	---
	b	11.3	---	---	---	---	--	---
TEST CONDITIONS:	a	6.3	---	165	---	0	--	---
	b	12.6	---	165	---	0	--	---

ABSOLUTE RATINGS:

	TYPE				
Parameter:	a,b	Rl	Cl	i surge/a	Alt
Unit:	a,b	Ohms	μf	ma	---
Maximum:	a	-----	--	176	See
	b	-----	--	---	Note 1
Minimum:	a	-----	--	---	---
	b	-----	--	---	---
TEST CONDITIONS:	a	10,000	8	---	---
	b	10,000	8	---	-------

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS Min	LIMITS Max	UNIT
	Qualification								
1216	Base material insulation quality	a, b	Min zone 5	---	--	----	---	---	----
1031	Variable-frequency vibration	a, b	No voltages	---	--	----	---	---	----
	Quality conformance inspection, part 1								
1353	Operation of rectifiers	a, b	See notes 2 and 5	0.65	II	Io	16.0	---	mAdc
1231	Emission	a, b	E2b = 0; E1b = 20 Vdc	0.65	II	I1s	15	---	mAdc
1231	Emission	a, b	E1b = 0; E2b = 20 Vdc	0.65	II	I2s	15	---	mAdc
1201	Shorts and discontinuity detection	a, b		0.4	II	----	---	---	----
	Quality conformance inspection, part 2								
1101	Secureness of base, cap or insert	a, b		---	--	----	---	---	----
1105	Permanence of marking	a, b		---	--	----	---	---	----
1111	Base pin solder depth	a, b		---	--	----	---	---	----
1211	Insulation of electrodes	a, b		---	--	----	---	---	----
1301	Heater current	a		2.5	S3	If	275	325	mA
1301	Heater current	b		2.5	S3	If	138	162	mA
1336	Heater-cathode leakage	a, b		2.5	S3	Ih1k Ih2k	--- ---	5 5	µAdc µAdc
1256	Electrode current	a, b	E1b = E2b = 0; R1 = 34,000 See note 3	6.5	S3	Ib	4.0	25.0	µAdc
1331	Direct interelectrode capacitance	a, b	1p to 1k, h and shell 2p to 2k, h and shell	6.5	Code E	C1p2p C C	--- 2.2 2.5	0.10 3.8 4.5	pF pF pF
	Quality conformance inspection, part 3								
1501	Intermittent life test (500 hours)	a, b	Group A See note 4	---	--	----	---	---	----
----	Intermittent life test end point: (500 hours)								
----	Operation of rectifiers	a, b		---	--	Io	14.0	---	mAdc

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NOTES:

1. Paragraph 6.5.2.3 of MIL-E-1G shall apply.
2. In a full-wave circuit, adjust Zp/a such that a tube having Etd = 11 Vdc at 16 mAdc per anode gives Io = 18 mAdc. Ehk = Eo + 117 Vac so phased that Ef and 117 subtract. Epp/a is loaded anode voltage.
3. Connect sections in parallel.
4. In life-test operation the values of R1 and C1 given in test conditions may be considered as approximate, and shall be adjusted initially to give Io equal to or greater than 18 mAdc; ib equal to, or greater than, 50 ma; epx equal to, or greater than, 465 v. Ehk = Eo + 117 Vac so phased that Ef and 117 Vac subtract. Epp/a is loaded anode voltage.
5. This test to be performed at the conclusion of the holding period.

Custodians:

Army - EL
Navy - EC
Air Force - 80

Preparing activity:

Air Force - 80

Review activities:

Army - EL
Navy - EC
Air Force - 17

(Project 5960-2651)

User activities:

Army - MU
Navy - WP, MC, CG
Air Force - 11

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